

# Lecture Scribe Guidelines



Create a formal lecture scribe using the context that is presented only. The document will be my only reference resource to use in preparing closed-notes exams.

To make sure that the scribe satisfies all elements of academic requirements, use a step-by-step Chain-of-Thought methodology when writing and directly provide your clear rationale in the final product. For the scribe you must follow the following constraints strictly:

**Clear Explanations:** New definitions, terms and assumptions. Take everything that the lectures introduce and explicitly describe it as it is introduced in the lecture.

**Step-by-Step Execution:** Trace the reasoning stepwise in all proofs, derivations, computations, and worked examples. Summarizing and omitting intermediate mathematical steps and simplifying the process should be avoided.

**Visible Reasoning:** Recurrently uncover logical connections amid dissimilar ideas in order to have the sequence of ideas clear on exam revision.

**Absolute Context Adherence:** Use pure materials given completely. Do not bring in external knowledge, different practices, unmotivated motives, and new models.

**Tone and Structure:** A formal academic. Divide the content

presented in a linear manner in the lecture using clear section titles.

The ultimate result should be a knowledgeable, reason-discovery writing of the lecture intended solely to study the exam.

Below is a **formal, exam-oriented lecture scribe** strictly based on the uploaded lecture:



All definitions, derivations, and examples are reconstructed **step-by-step with explicit reasoning**, without adding any external material.

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## Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

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### 1. Marginal PMF

#### 1.1 Definition

For two discrete random variables  $X$  and  $Y$ , the **joint PMF** is defined as:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

#### Key Idea (from lecture)

The **marginal PMF of one variable** is obtained by **summing the joint PMF over all possible values of the other variable**.

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#### 1.2 Step-by-Step Derivation

To extract marginal distributions:

For  $X$ :

$$p_X(x) = \sum_y p_{XY}(x, y)$$

Reasoning:

- Fix  $x$
  - Sum probabilities over all possible  $y$
  - This accounts for all outcomes where  $X = x$
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For  $Y$ :

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Reasoning:

- Fix  $y$
  - Sum over all  $x$
  - This captures all events where  $Y = y$
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### 1.3 Marginal PMF from Joint PMF Table

From the table (page 4):

- Row sums  $\rightarrow p_X(x)$
- Column sums  $\rightarrow p_Y(y)$

Mathematically:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

**Normalization Property:**

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$


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### 1.4 Worked Example (Step-by-Step)

From the table (page 5):

| $X \backslash Y$ | $y = 0$ | $y = 1$ |
|------------------|---------|---------|
| $x = 0$          | $1/8$   | $1/4$   |
| $x = 1$          | $3/8$   | $1/4$   |

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**Step 1: Compute  $p_X(x)$**

For  $x = 0$ :

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$

Convert to common denominator:

$$= \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

For  $x = 1$ :

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$


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**Step 2: Compute  $p_Y(y)$**

For  $y = 0$ :

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

For  $y = 1$ :

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

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**Final Marginals**

$$p_X(x) = \begin{cases} 3/8, & x = 0 \\ 5/8, & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2, & y = 0 \\ 1/2, & y = 1 \end{cases}$$

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## 2. Correlation

### 2.1 Definition

Correlation is introduced via the **product moment**:

$$R_{XY} = E[XY]$$

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### 2.2 Interpretation (Step-by-Step)

From lecture (pages 6–7):

- Measures **average value of product  $XY$**
- Depends on:
  - Mean values
  - Spread
  - Dependence

## Important Observation

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

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## 2.3 Limitation

- Computed about the **origin (not means)**
- Mixes:
  - mean
  - variance
  - dependence

➡ Therefore, it is **not a clean measure of dependence**

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## Transition to Covariance

To remove mean effects:

$$\text{Center variables} \Rightarrow \text{Covariance}$$

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## 3. Covariance

### 3.1 Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

### Step-by-Step Meaning

1. Subtract means:
  - $X - \mu_X$
  - $Y - \mu_Y$
2. Multiply deviations:

- Measures joint variation
3. Take expectation:
- Average behavior
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### 3.2 Interpretation

From page 9:

- $\text{Cov}(X, Y) > 0$ : same direction
- $\text{Cov}(X, Y) < 0$ : opposite direction
- $\text{Cov}(X, Y) = 0$ : no linear relation

⚠ Magnitude depends on units → cannot measure strength directly

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### 3.3 Algebraic Derivation (Step-by-Step)

Start with:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand product

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply expectation linearly

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Substitute means

$$\begin{aligned} E[X] &= \mu_X, & E[Y] &= \mu_Y \\ &= E[XY] - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y \\ &= E[XY] - \mu_X\mu_Y \end{aligned}$$

### Final Formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

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### 3.4 Properties

#### 1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

#### 2. Variance case

$$\text{Cov}(X, X) = \text{Var}(X)$$

#### 3. Linearity

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

#### 4. Independence

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

⚠ Important:

Zero covariance  $\neq$  independence (general case)

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### 3.5 Example 1

Given:

$$Y = -2X + 3$$

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#### Step 1: Formula



$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

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Step 2: Compute  $XY$

$$XY = X(-2X + 3) = -2X^2 + 3X$$

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Step 3: Compute  $E[Y]$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

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Step 4: Substitute

$$\begin{aligned}\text{Cov}(X, Y) &= E[-2X^2 + 3X] - E[X](-2E[X] + 3) \\ &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X] \\ &= -2E[X^2] + 2(E[X])^2\end{aligned}$$

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Step 5: Use variance

$$\begin{aligned}\text{Var}(X) &= E[X^2] - (E[X])^2 \\ \Rightarrow \text{Cov}(X, Y) &= -2\text{Var}(X)\end{aligned}$$

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## 4. Pearson's Correlation Coefficient

### 4.1 Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

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## 4.2 Interpretation

- Measures:
  - **Direction**
  - **Strength**
- Properties:
  - Unit-free
  - Normalized covariance
  - Range:

$$-1 \leq \rho_{XY} \leq 1$$

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## 4.3 Meaning

- $\rho > 0$ : positive linear trend
  - $\rho < 0$ : negative linear trend
  - $\rho = 0$ : no linear correlation
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# 5. Joint Gaussian Random Variables

## 5.1 Definition

A pair  $(X, Y)$  is **jointly Gaussian** if:

- Joint PDF has **bivariate Gaussian form**
  - Marginals are Gaussian
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## 5.2 Algebraic Form

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[ \left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

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## 5.3 Parameters

- $\mu_X, \mu_Y$ : means
  - $\sigma_X, \sigma_Y$ : standard deviations
  - $\rho$ : correlation coefficient
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## 5.4 Interpretation (from lecture example)

Example (page 23):

- $X$ : CO<sub>2</sub> concentration
- $Y$ : temperature

If:

$$\rho > 0$$

➡ Higher CO<sub>2</sub> → higher temperature

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## 5.5 Applications

- Signal processing
  - Communication systems
  - Modeling noisy measurements
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## Final Concept Flow (Important for Exam)

1. Joint PMF  $\rightarrow$  Marginal PMF (via summation)
  2. Product moment  $E[XY] \rightarrow$  raw correlation
  3. Centering  $\rightarrow$  Covariance
  4. Normalize  $\rightarrow$  Pearson coefficient
  5. Continuous extension  $\rightarrow$  Joint Gaussian
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If you want, I can convert this into a **perfect 1-page revision sheet** or **MCQ-focused notes** for your exam.